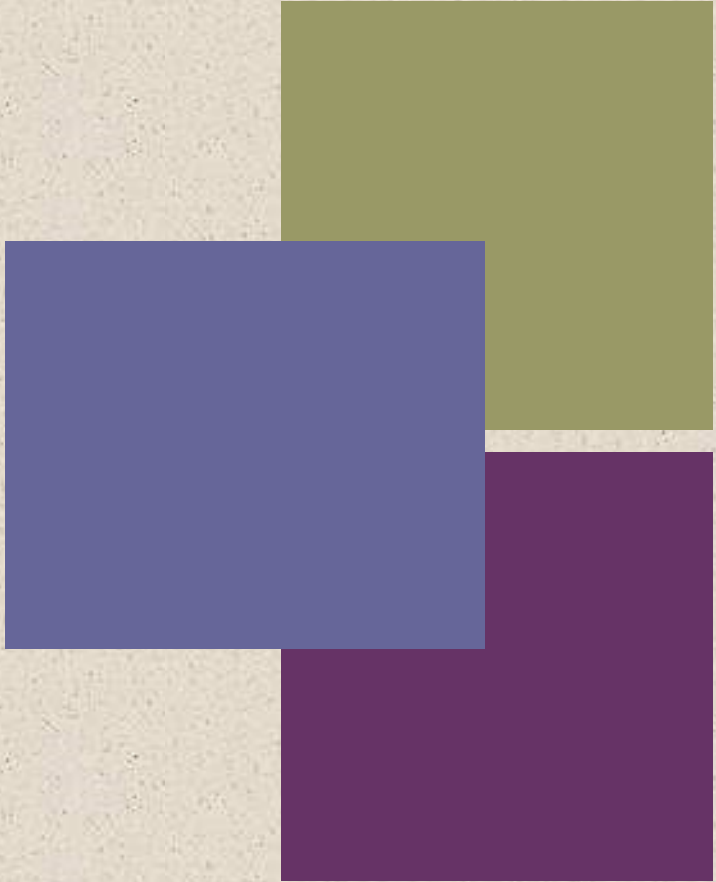




# + Chapter 17

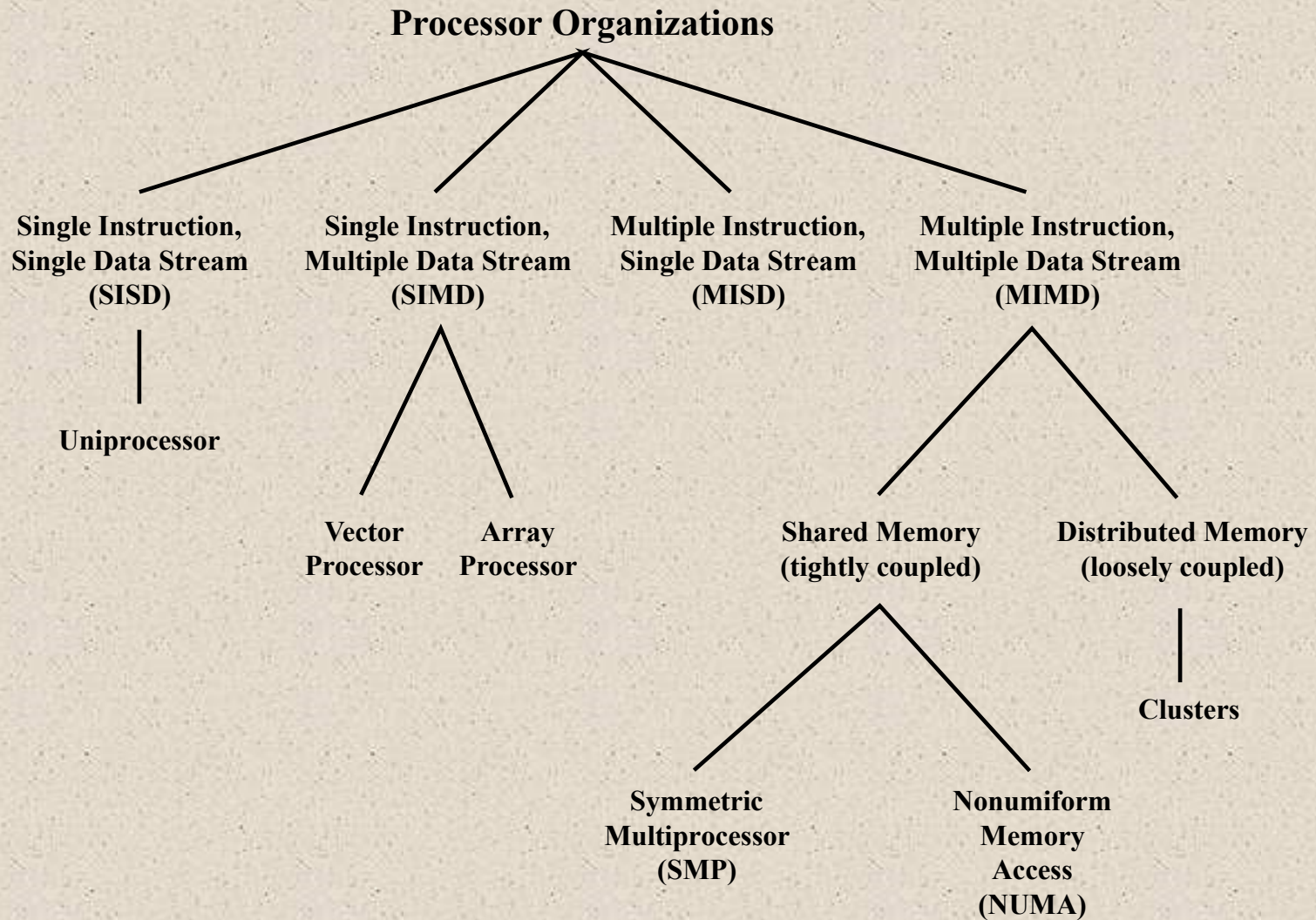
## Parallel Processing



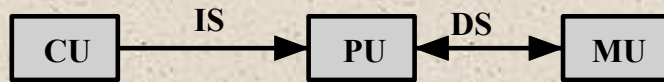
# Multiple Processor Organization



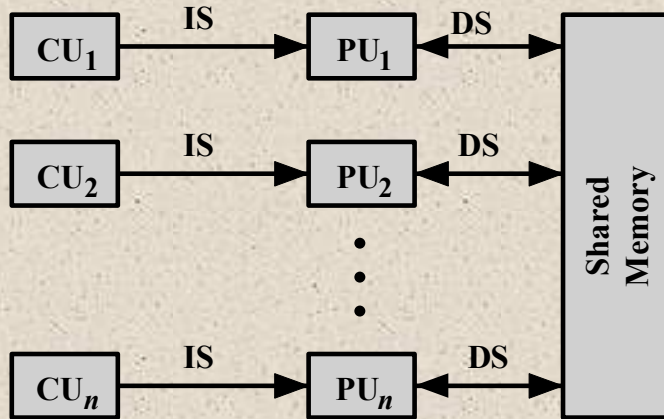
- Single instruction, single data (**SISD**) stream
  - Single processor executes a single instruction stream to operate on data stored in a single memory
  - Uniprocessors fall into this category
- Single instruction, multiple data (**SIMD**) stream
  - A single machine instruction controls the simultaneous execution of a number of processing elements on a lockstep basis
  - Vector and array processors fall into this category
- Multiple instruction, single data (**MISD**) stream
  - A sequence of data is transmitted to a set of processors, each of which executes a different instruction sequence
  - Not commercially implemented
- Multiple instruction, multiple data (**MIMD**) stream
  - A set of processors simultaneously execute different instruction sequences on different data sets
  - SMPs, clusters and NUMA systems fit this category



**Figure 17.1 A Taxonomy of Parallel Processor Architectures**

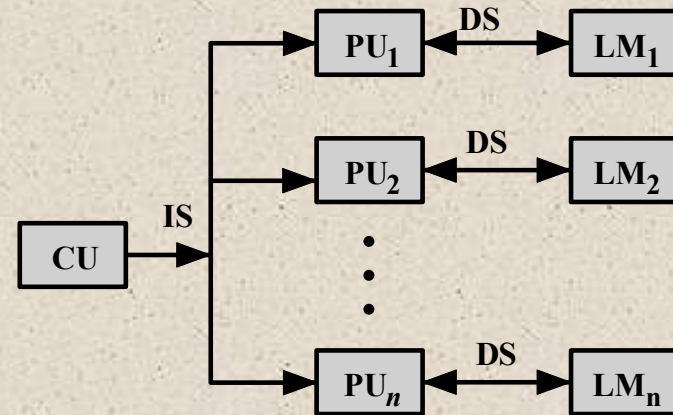


(a) SISD

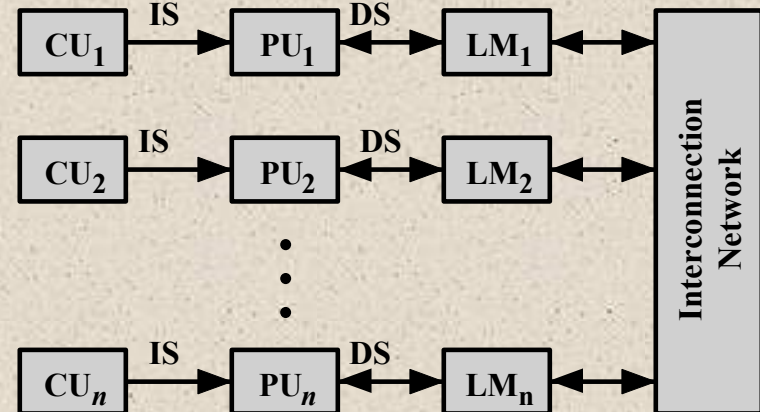


(c) MIMD (with shared memory)

|                         |                                                      |
|-------------------------|------------------------------------------------------|
| CU = control unit       | SISD = single instruction,<br>single data stream     |
| IS = instruction stream | SIMD = single instruction,<br>multiple data stream   |
| PU = processing unit    | MIMD = multiple instruction,<br>multiple data stream |
| DS = data stream        |                                                      |
| MU = memory unit        |                                                      |
| LM = local memory       |                                                      |



(b) SIMD (with distributed memory)



(d) MIMD (with distributed memory)

**Figure 17.2 Alternative Computer Organizations**

# Symmetric Multiprocessor (SMP)



A stand alone computer with the following characteristics:

|                                                       |                                                                                                                                                                                                                                                   |                                                                                                                                                                                     |                                                                          |                                                                                                                                                                                                                 |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Two or more similar processors of comparable capacity | <p>Processors share same memory and I/O facilities</p> <ul style="list-style-type: none"><li>• Processors are connected by a bus or other internal connection</li><li>• Memory access time is approximately the same for each processor</li></ul> | <p>All processors share access to I/O devices</p> <ul style="list-style-type: none"><li>• Either through same channels or different channels giving paths to same devices</li></ul> | <p>All processors can perform the same functions (hence “symmetric”)</p> | <p>System controlled by integrated operating system</p> <ul style="list-style-type: none"><li>• Provides interaction between processors and their programs at job, task, file and data element levels</li></ul> |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|





(a) Interleaving (multiprogramming, one processor)



(b) Interleaving and overlapping (multiprocessing; two processors)

Blocked Running

**Figure 17.3 Multiprogramming and Multiprocessing**



# Multithreading and Chip Multiprocessors



## ■ Multithreading

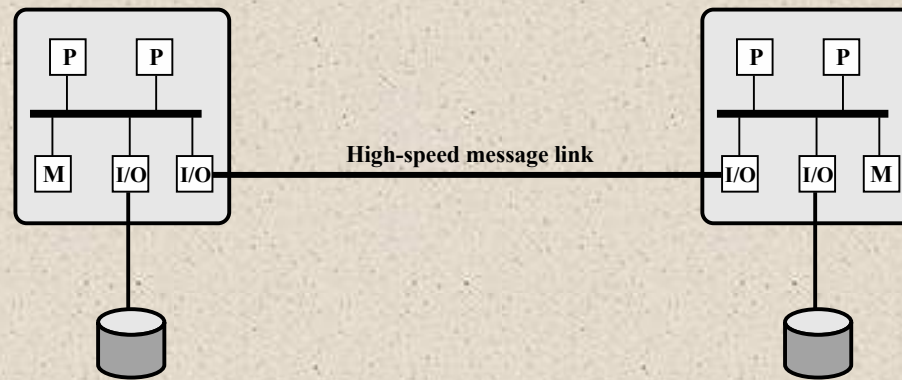
- Allows for a high degree of instruction-level parallelism without increasing circuit complexity or power consumption
- Instruction stream is divided into several smaller streams, known as threads, that can be executed in parallel

# Clusters

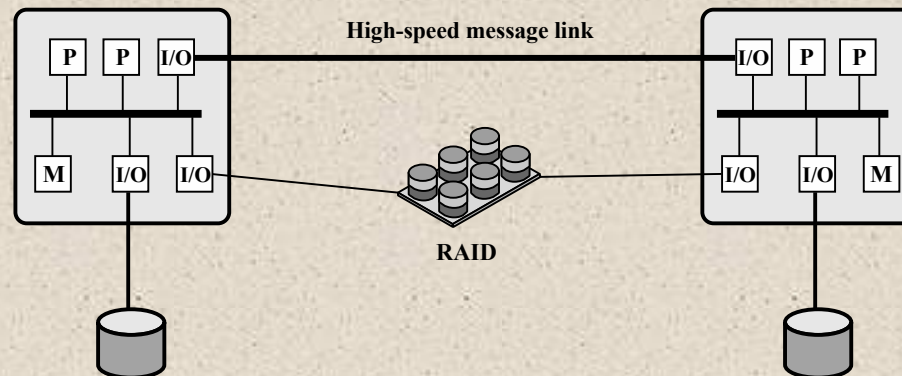
- Alternative to SMP as an approach to providing high performance and high availability
- Particularly attractive for server applications
- Defined as:
  - A group of interconnected whole computers working together as a unified computing resource that can create the illusion of being one machine
  - (The term *whole computer* means a system that can run on its own, apart from the cluster)
- Each computer in a cluster is called a node
- Benefits:
  - Absolute scalability
  - Incremental scalability
  - High availability
  - Superior price/performance







(a) Standby server with no shared disk



(b) Shared disk

Figure 17.8 Cluster Configurations



# Operating System Design Issues



- How failures are managed depends on the clustering method used
- Two approaches:
  - Highly available clusters
  - Fault tolerant clusters
- Failover
  - The function of switching applications and data resources over from a failed system to an alternative system in the cluster
- Failback
  - Restoration of applications and data resources to the original system once it has been fixed
- Load balancing
  - Incremental scalability
  - Automatically include new computers in scheduling
  - Middleware needs to recognize that processes may switch between machines



# Clusters Compared to SMP

- Both provide a configuration with multiple processors to support high demand applications
- Both solutions are available commercially

## SMP

- Easier to manage and configure
- Much closer to the original single processor model for which nearly all applications are written
- Less physical space and lower power consumption
- Well established and stable

## Clustering

- Far superior in terms of incremental and absolute scalability
- Superior in terms of availability
- All components of the system can readily be made highly redundant



# Nonuniform Memory Access (NUMA)



- Alternative to SMP and clustering
- Uniform memory access (UMA)
  - All processors have access to all parts of main memory using loads and stores
  - Access time to all regions of memory is the same
  - Access time to memory for different processors is the same
- Nonuniform memory access (NUMA)
  - All processors have access to all parts of main memory using loads and stores
  - Access time of processor differs depending on which region of main memory is being accessed
  - Different processors access different regions of memory at different speeds
- Cache-coherent NUMA (CC-NUMA)
  - A NUMA system in which cache coherence is maintained among the caches of the various processors

# Motivation

SMP has practical limit to number of processors that can be used

- Bus traffic limits to between 16 and 64 processors

In clusters each node has its own private main memory

- Applications do not see a large global memory
- Coherency is maintained by software rather than hardware

NUMA retains SMP flavor while giving large scale multiprocessing

Objective with NUMA is to maintain a transparent system wide memory while permitting multiple multiprocessor nodes, each with its own bus or internal interconnect system



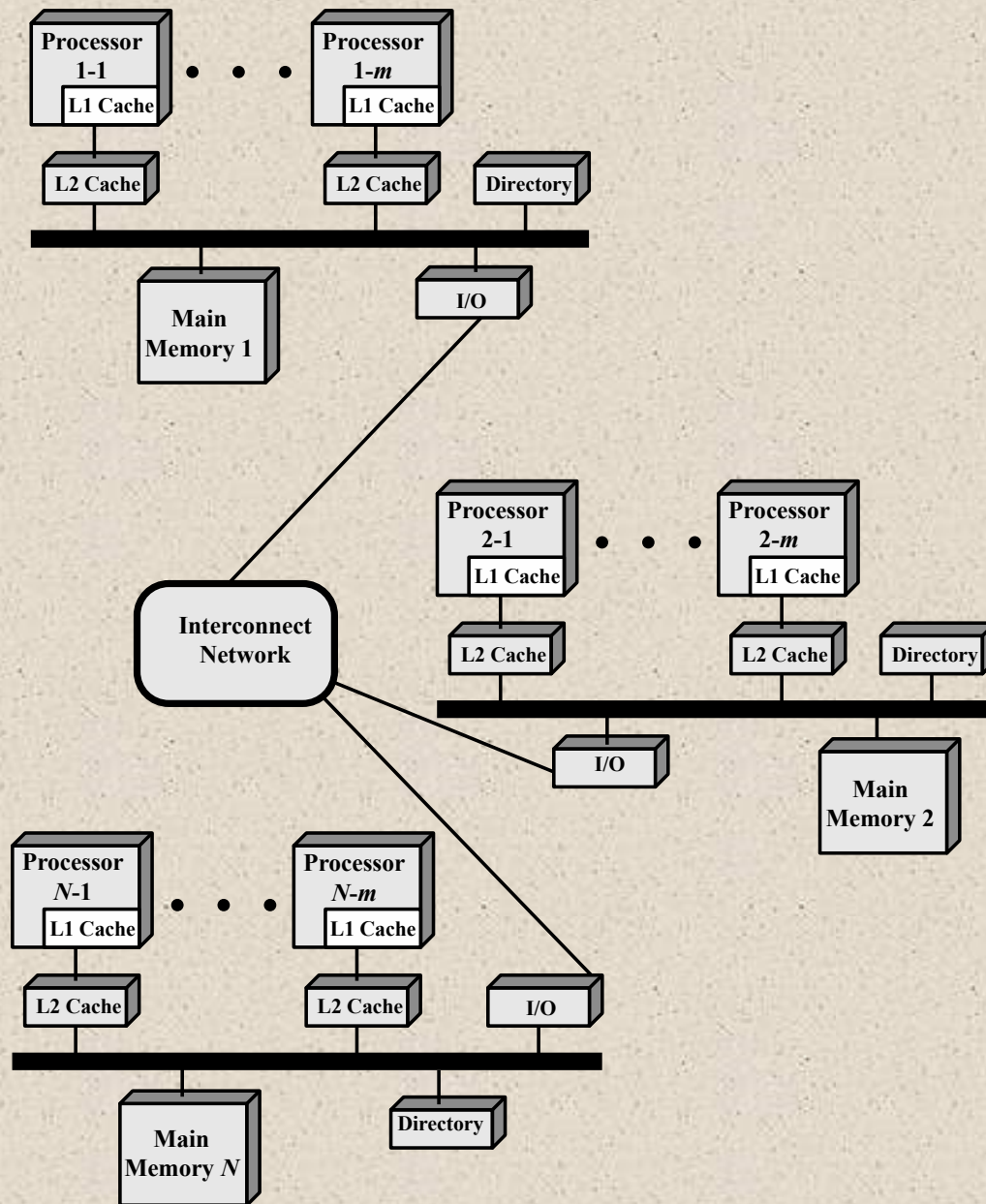


Figure 17.11 CC-NUMA Organization



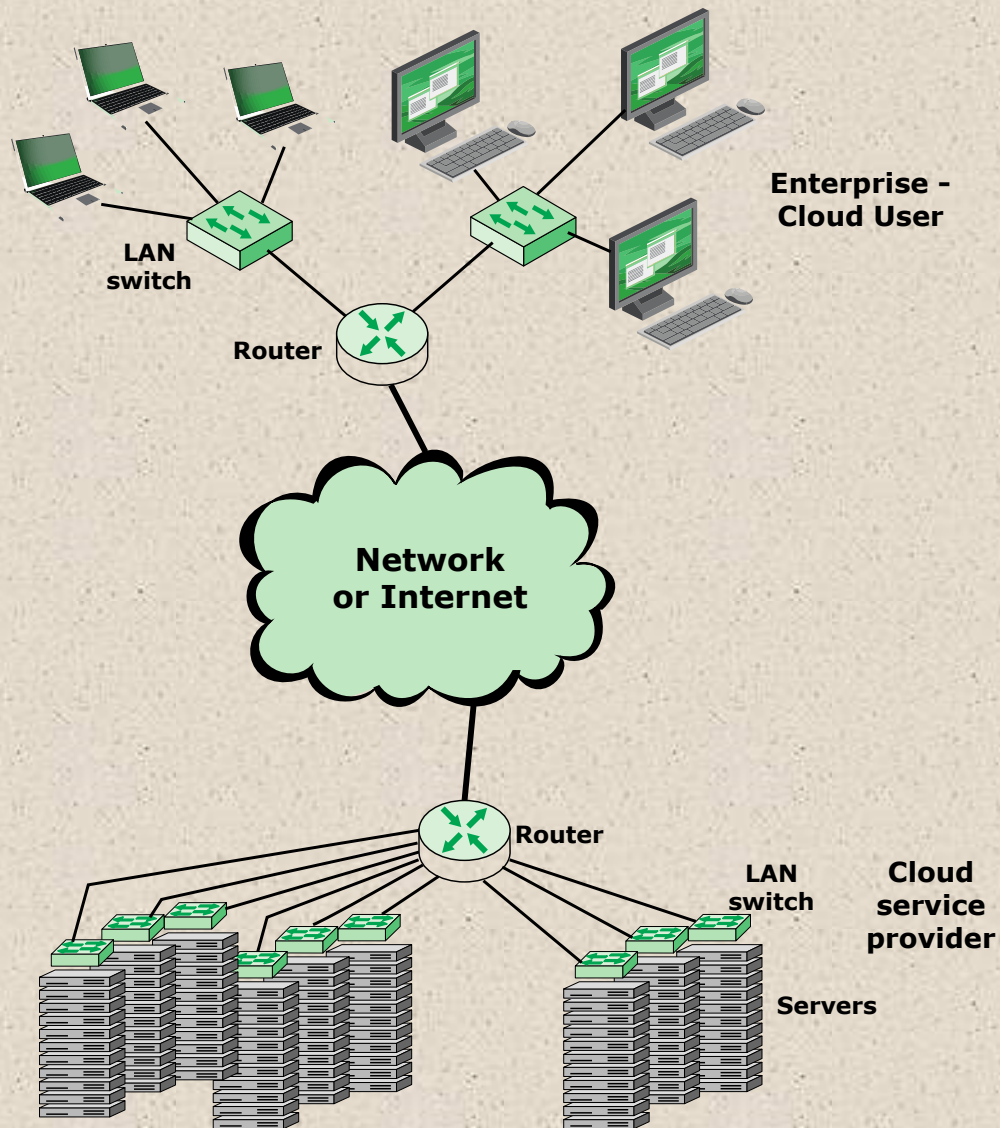
# NUMA Pros and Cons



- Main advantage of a CC-NUMA system is that it can deliver effective performance at higher levels of parallelism than SMP without requiring major software changes
- Bus traffic on any individual node is limited to a demand that the bus can handle
- If many of the memory accesses are to remote nodes, performance begins to break down
- Does not transparently look like an SMP
- Software changes will be required to move an operating system and applications from an SMP to a CC-NUMA system
- Concern with availability

# + Deployment Models

- Public cloud
  - The cloud infrastructure is made available to the general public or a large industry group and is owned by an organization selling cloud services
  - Major advantage is cost
- Private cloud
  - A cloud infrastructure implemented within the internal IT environment of the organization
  - A key motivation for opting for a private cloud is security
- Community cloud
  - Like a private cloud it is not open to any subscriber
  - Like a public cloud the resources are shared among a number of independent organizations
- Hybrid cloud
  - The cloud infrastructure is a composition of two or more clouds that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability
  - Sensitive information can be placed in a private area of the cloud and less sensitive data can take advantage of the cost benefits of the public cloud



**Figure 17.14 Cloud Computing Context**